

One-Page Summary – Social Media Engagement Analytics

This project analysed a social media engagement dataset containing 5,000 records and 19 original variables. The analysis focused on understanding patterns in likes, comments, shares, watch time, impressions, engagement rate, user characteristics, content categories, device usage and sentiment. The data was cleaned by identifying missing values, filling numerical missing values using the median, handling categorical missing values, standardising gender labels, checking duplicate records and creating a new hashtag count variable. No duplicate records were found, and the dataset was complete after the cleaning process.

The descriptive analysis showed that the average number of likes was 10,106.98, average comments were 1,502.04, average shares were 1,002.91, and average watch time was 4,014.50 seconds. The average engagement rate was 0.9644, while the average follower count was approximately 393,698. The correlation analysis showed generally weak relationships among the numerical variables, indicating that no strong linear relationship was observed between most engagement-related variables.

For content performance, image posts recorded the highest average engagement score at approximately 16,213.19, followed by text, video and reel posts. The food category was the best-performing category based on average likes, with approximately 10,225.50 likes, and it also recorded the highest average engagement rate of 1.3586. At the country level, Brazil recorded the highest average engagement rate at approximately 1.5407.

The user trends analysis showed differences in engagement across age groups. The 46–55 age group had the highest average engagement rate of approximately 1.4141, while users above 55 had the lowest at approximately 0.7092. Verified accounts also performed better than non-verified accounts, with average engagement rates of 1.0543 and 0.9547, respectively.

The behavioural analysis found that mobile users had the highest average watch time at approximately 4,087.83 seconds, compared with 3,979.74 seconds for tablets and 3,974.79 seconds for desktops. The best time of day for impressions could not be meaningfully identified because all records were recorded at hour 0 after converting the date field to datetime.

Finally, sentiment analysis showed that negative posts had the highest average engagement rate at 1.0385, followed by neutral posts at 0.9912 and positive posts at 0.9197. Negative posts also had the highest average likes, while neutral posts had the highest average comments. Overall, the analysis indicates that content category, post type, country, age, verification status, device type and sentiment are associated with differences in social media engagement. These findings can help identify the types of content and user characteristics associated with stronger engagement in the analysed dataset.